

Practical No.3

Aim: Create a program to demonstrate encapsulation by defining private fields and accessing them through public get and set methods.

Source code:

```
public class Employee {  
    private String name;  
    private int age;  
    private String department;  
    public Employee(String name, int age, String department) {  
        this.name = name;  
        this.age = age;  
        this.department = department;  
    }  
    public String getName() {  
        return this.name;  
    }  
    public int getAge() {  
        return this.age;  
    }  
    public String getDepartment() {  
        return this.department;  
    }  
    public void setName(String name) {  
        this.name = name;  
    }  
    public void setAge(int age) {  
        if (age > 0) {  
            this.age = age;  
        } else {  
            System.out.println("Invalid age. Age should be greater than 0.");  
        }  
    }  
    public void setDepartment(String department) {  
        this.department = department;  
    }  
}
```

```
}  
public static void main(String[] args) {  
    Employee employee = new Employee("Vinay V", 30, "CS");  
  
    System.out.println("Name: " + employee.getName());  
    System.out.println("Age: " + employee.getAge());  
    System.out.println("Department: " + employee.getDepartment());  
  
    employee.setName("Shubham D");  
    employee.setAge(19);  
    employee.setDepartment("MCA");  
  
    System.out.println("\nAfter Updating Details:");  
    System.out.println("Updated Name: " + employee.getName());  
    System.out.println("Updated Age: " + employee.getAge());  
    System.out.println("Updated Department:" + employee.getDepartment());  
}  
}
```

Output:

```
Name: Vinay V  
Age: 30  
Department: CS
```

```
After Updating Details:  
Updated Name: Shubham D  
Updated Age: 19  
Updated Department:MCA
```